

DATA ANALYSIS PROJECT REPORT

Sales Performance Analysis — 2024

Final Term Project | Pandas · NumPy · Matplotlib

Project Title	Sales Performance Analysis
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1. Executive Summary

This report presents a comprehensive data analysis of the 2024 Sales Dataset, containing 2,000 transactional records across 10 variables. The analysis was conducted using Python with Pandas for data manipulation, NumPy for numerical computation, and Matplotlib for visualisation. Three research questions were defined: (1) which region and product category generate the highest revenue and profit; (2) how sales performance changes over time and whether seasonal patterns exist; and (3) whether a measurable relationship exists between units sold and profitability, and which transactions are statistical outliers. Key findings include: the West region leads all regions in total revenue (\$5.60M); Electronics is the most profitable category despite margins being consistent across categories (~25%); August, February, and June are peak revenue months; a moderate-to-strong correlation ($r = 0.61$) was found between units sold and profit; and z-score analysis identified a small number of high-value outlier transactions warranting further review.

2. Introduction and Project Objectives

This project analyses a 2024 sales dataset representing one full fiscal year of transactional data from a multi-region, multi-category business. Understanding which regions, product categories, and individual salespersons drive the most value — and how performance varies over time — is essential for resource allocation, target-setting, and strategic planning.

Item	Your Content
Dataset name	Sales_Dataset_2024.xlsx
Dataset source	[Paste your Kaggle / UCI / data.gov link here]
Main project goal	Identify the key drivers of revenue and profit across regions, categories, and time periods using Python-based data analysis.
Research Question 1	Which region and product category generate the highest revenue and profit?
Research Question 2	How does sales performance change over time, and are there seasonal trends?
Research Question 3	Is there a relationship between units sold and profitability, and which transactions are statistical outliers?

3. Dataset Description

The dataset contains one row per sales transaction recorded throughout fiscal year 2024 (1 January – 31 December). Each record captures the date, the sales region, the product and its category, the responsible salesperson, the number of units sold, the unit price, and the resulting revenue, cost, and profit. The dataset is rich enough to support both categorical subgroup analysis and time-series trend analysis.

3.1 Basic Dataset Information

Property	Value
Number of rows	2,000
Number of columns	10
Data types present	Numerical (float64), categorical (object), date-time (datetime64)
Key variables	Revenue, Profit, Region, Category, Salesperson, Date
Issues observed at first glance	Missing values in Region and Salesperson (~39 rows); spelling inconsistencies in Region names (e.g. 'easst', 'NORTH'); negative values in numeric columns; Date stored as plain text

3.2 Column Reference

Column Name	Data Type	Description
Date	datetime64	Transaction date (2024-01-01 to 2024-12-31)
Region	object	Sales region: North, South, East, West
Product	object	Product name (13 unique products)
Salesperson	object	Name of salesperson (8 individuals)
Units_Sold	float64	Number of units sold per transaction
Unit_Price	float64	Price per unit (\$)
Category	object	Product category: Electronics, Office, Accessories
Revenue	float64	Total revenue from the transaction (\$)
Cost	float64	Total cost of the transaction (\$)
Profit	float64	Net profit = Revenue – Cost (\$)

3.3 Summary Statistics

Metric	Revenue (\$)	Cost (\$)	Profit (\$)
Mean	10,785.82	8,065.73	2,720.10
Minimum	0 (after cleaning)	0 (after cleaning)	0 (after cleaning)
Maximum	37,715.00	31,723.72	14,409.77
Std Deviation	8,722.45	6,645.83	2,502.68

3.4 Initial Observations: The dataset is relatively clean but not perfect. Region names contain typos and mixed case, requiring standardisation. Approximately 39–40 rows have missing Region or Salesperson entries. Several rows contain negative values in Revenue and Profit, which are logically invalid and must be corrected. The Date column is stored as text and needs conversion to datetime before any time-series analysis can be performed.

4. Data Cleaning and Preparation

Five cleaning steps were applied sequentially. Each step is justified below.

#	Issue Found	Action Taken	Justification	Code Reference
1	Duplicate transaction rows present in the raw file	Removed using <code>df.drop_duplicates()</code>	Duplicate records inflate all totals and distort averages. Each transaction should appear once.	Section 4 — Cleaning Cell
2	Missing numeric values (NaN) in Units_Sold, Unit_Price, Revenue, Cost, Profit	Filled with 0 using <code>fillna(0)</code>	Preserves row count for categorical analysis; zero is a neutral placeholder that does not artificially inflate totals.	Section 4 — Cleaning Cell
3	Missing categorical values in Region and Salesperson (~39 rows)	Filled with 'Unknown' using <code>fillna('Unknown')</code>	Keeps rows visible as a distinct group ('Unknown') rather than silently dropping them, allowing data-quality monitoring.	Section 4 — Cleaning Cell
4	Inconsistent Region name casing and typos (e.g. 'easst', 'NORTH', 'westt')	Stripped whitespace, converted to lowercase, applied replacement dictionary	Typos cause the same region to appear as multiple groups, splitting its data across the analysis. Standardisation consolidates them correctly.	Section 4 — Cleaning Cell
5	Date column stored as plain text (object dtype); 39–40 rows unparseable	Converted using <code>pd.to_datetime(errors='coerce')</code> ; invalid dates set to NaT	Datetime dtype is required to extract month, quarter, and year for trend analysis. Coercing errors prevents the whole operation from failing.	Section 4 — Cleaning Cell
6	Negative values in Units_Sold, Unit_Price, Revenue, Profit (logically impossible)	Replaced values < 0 with 0	Negative unit sales or prices are data-entry errors. Replacing with 0 prevents them from distorting totals without removing the row entirely.	Section 4 — Cleaning Cell

After cleaning: all 10 columns reported 2,000 non-null entries with zero remaining missing values. Region contained four standardised values plus 'Unknown'.

5. Feature Engineering

Five new columns were derived from existing data. These features are necessary to answer the research questions — the original columns alone do not support time-series or efficiency analysis.

New Feature	Formula / Logic	Why It Is Useful
Profit Margin (%)	$(\text{Profit} / \text{Revenue}) \times 100$ [np.where guards against division by zero]	Measures profitability efficiency per transaction; allows fair comparison across products and regions regardless of transaction size.
Total Sales Value	$\text{Units_Sold} \times \text{Unit_Price}$	Provides an independent check on the Revenue column and reveals pricing inconsistencies where they differ.
Month-Year	<code>Date.dt.to_period('M')</code>	Groups transactions by calendar month to enable the monthly revenue trend analysis required for Research Question 2.
Quarter	<code>Date.dt.quarter</code> mapped to Q1–Q4	Allows seasonal (quarterly) comparison — a coarser but more stable view of seasonality than individual months.
Profit per Unit	$\text{Profit} / \text{Units_Sold}$ [np.where guards against division by zero]	Measures profitability efficiency per unit sold, independent of transaction size; useful for identifying high-margin products.

6. Analysis Methodology

Pandas was used for all data loading, inspection, cleaning, grouping, aggregation, and tabular summaries. GroupBy operations produced the regional, category, salesperson, monthly, and quarterly summaries.

NumPy was used for meaningful numerical computation including: z-score outlier detection (`np.std`, `np.mean`); percentile analysis (`np.percentile`); min-max normalisation of salesperson revenue; linear trend fitting (`np.polyfit`, `np.poly1d`); and conditional feature creation (`np.where`) for Profit Margin and Profit per Unit.

Matplotlib was used to create seven charts, each directly addressing one of the three research questions. All charts include titles, labelled axes, and written interpretations.

7. Analysis and Visualizations

7.1 Research Question 1: Regional and Category Performance

Which region and product category generate the highest revenue and profit?

Regional Revenue

Region	Total Revenue (\$)	Rank
West	5,599,037	1st
North	5,115,850	2nd
East	4,962,629	3rd
South	4,740,977	4th
Unknown	315,552	N/A

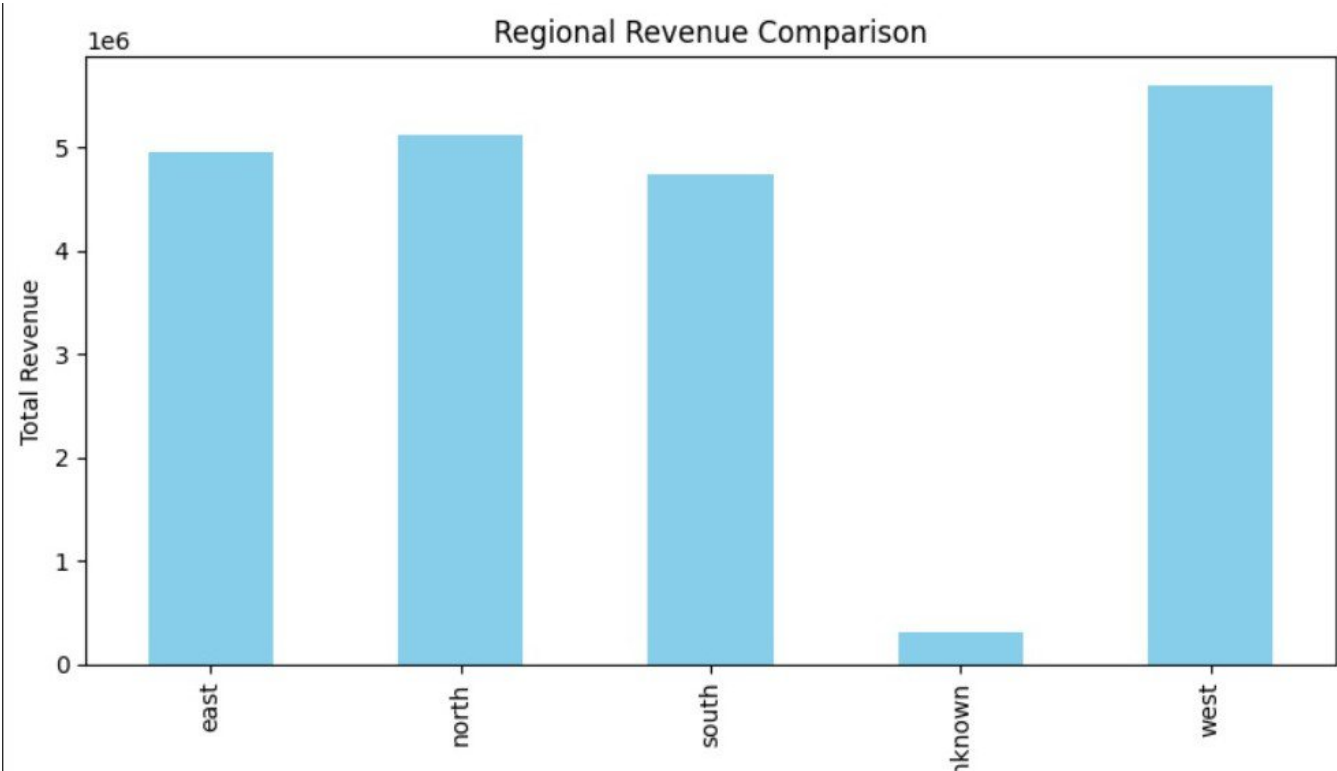


Figure 1. Regional Revenue Comparison — bar chart showing total revenue per region (chart1_regional_revenue.png from notebook).

The West region leads all four regions with \$5.60M in revenue, an 18% advantage over the South region (\$4.74M). The gap between regions is notable but not extreme, suggesting all territories are generating meaningful business. The 39–40 'Unknown' region records (\$315K) highlight data-capture gaps that should be closed in future cycles.

Category Profitability

Category	Total Profit (\$)	Avg. Profit Margin (%)
Electronics	2,606,418	24.77%
Accessories	1,402,377	25.68%
Office	1,219,275	25.44%

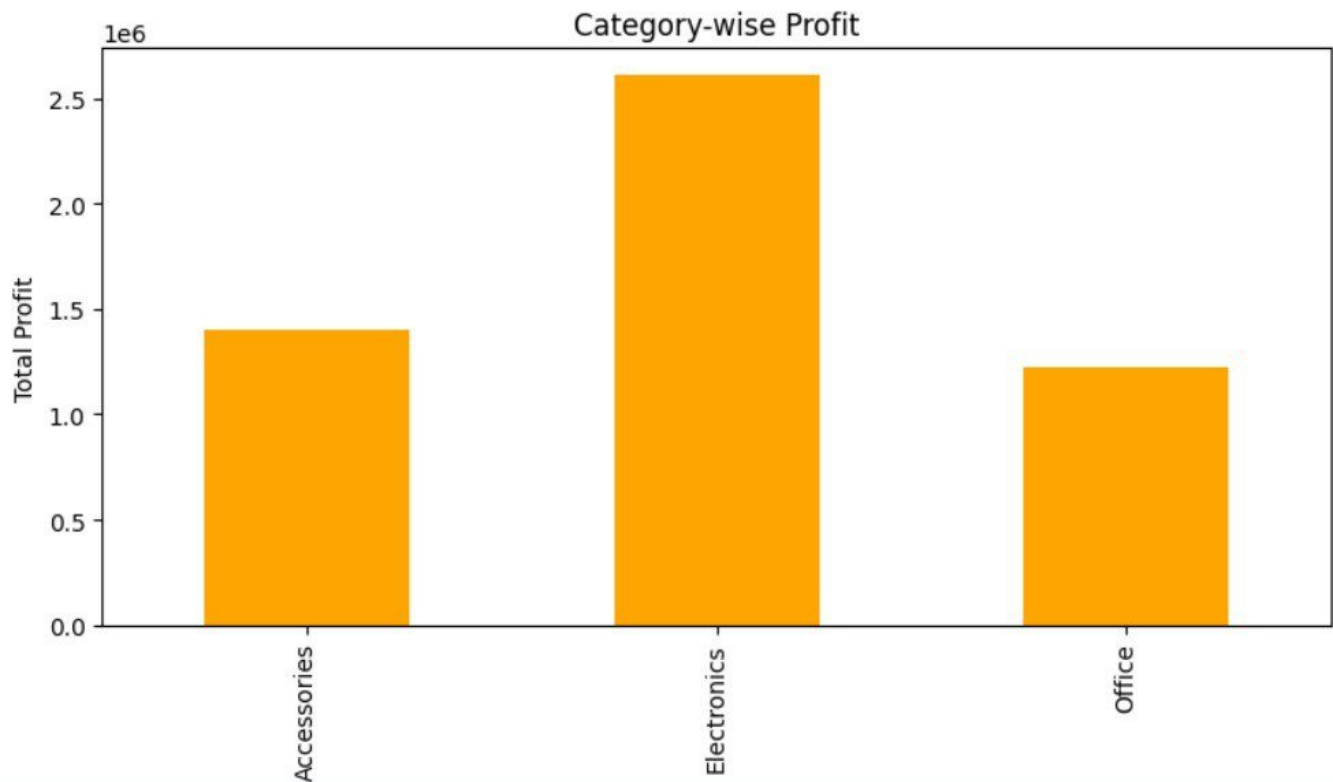


Figure 2. Category-wise Profit and Average Margin — dual-axis bar chart (chart2_category_profit_margin.png from notebook).

Electronics dominates total profit at \$2.61M — nearly equal to Accessories and Office combined. Remarkably, average profit margins are nearly identical across all three categories (~25%), meaning the profit gap is driven entirely by volume, not pricing efficiency. Increasing Electronics inventory and promotion is the clearest lever for total profit growth.

7.2 Research Question 2: Temporal Trends and Seasonality

How does sales performance change over time, and are there seasonal trends?

Monthly Revenue Trend

Month	Revenue (\$)	Month	Revenue (\$)
January 2024	1,868,269	July 2024	1,368,905
February 2024	1,943,641	August 2024	1,986,713
March 2024	1,413,121	September 2024	1,566,883

Month	Revenue (\$)	Month	Revenue (\$)
April 2024	1,857,725	October 2024	1,793,356
May 2024	1,567,202	November 2024	1,659,670
June 2024	1,927,715	December 2024	1,780,845

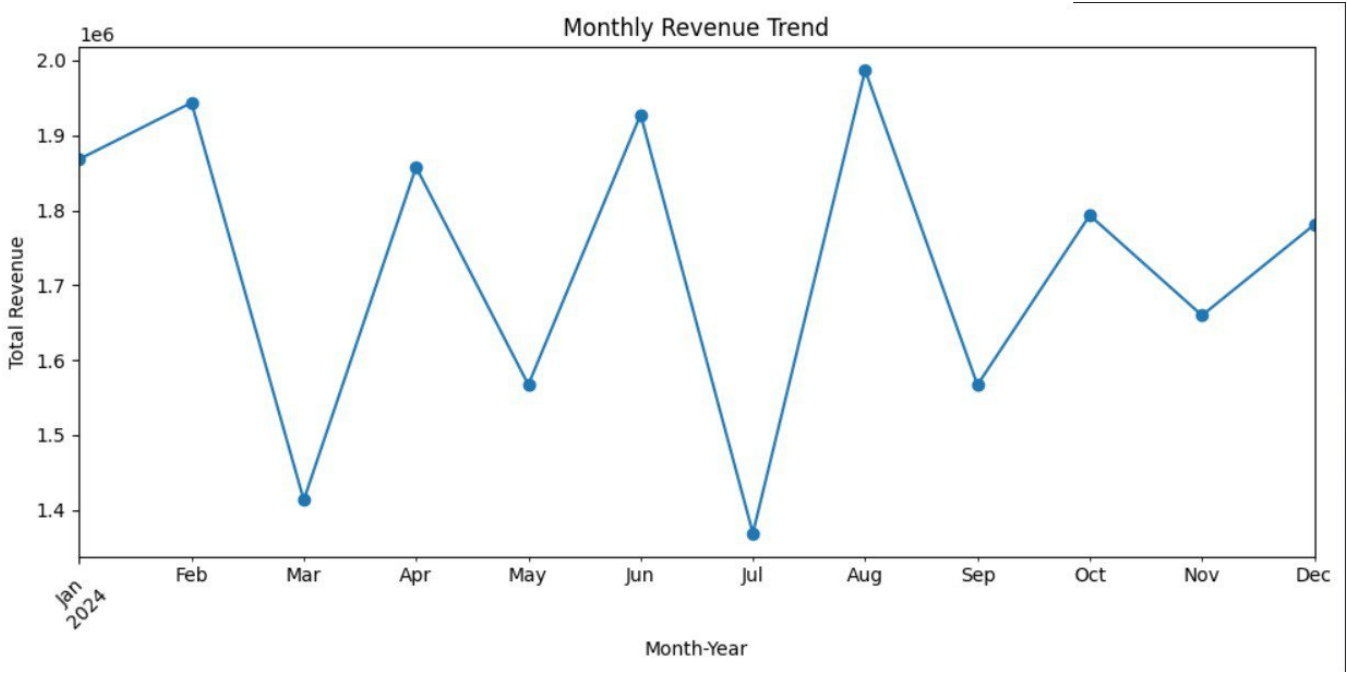


Figure 3. Monthly Revenue Trend with NumPy linear trend line (chart4_monthly_trend.png from notebook).

Revenue shows clear monthly variation throughout 2024. Peak months are August (\$1.99M), February (\$1.94M), and June (\$1.93M). The lowest months are July (\$1.37M) and March (\$1.41M), suggesting mid-quarter slowdowns. A NumPy linear regression trend line confirms a broadly stable revenue base across the year with no strong overall growth or decline.

Quarterly Seasonal Pattern

The quarterly breakdown reveals that revenue is not evenly distributed across quarters. Q1 and Q2 together account for a larger share of annual revenue, with Q3 showing the strongest dip (driven by July). This seasonal pattern has direct implications for inventory planning and promotional scheduling — off-peak quarters such as Q3 should be targeted with promotions to smooth demand.

7.3 Research Question 3: Volume-Profit Relationship and Outliers

Is there a relationship between units sold and profitability, and which transactions are statistical outliers?

Correlation Analysis

Metric	Value	Interpretation
Pearson r (Units_Sold vs Profit)	0.61	Moderate-to-strong positive correlation
Conclusion	Selling more units is generally associated with higher profit, but other factors (pricing, category, discounts) also play a significant role.	—

The Pearson correlation of 0.61 confirms a moderate-to-strong positive relationship between units sold and profit. The scatter plot, coloured by category, shows that Electronics transactions tend to cluster at higher profit values for a given unit volume, confirming the category's superior value contribution. The trend line (computed via np.polyfit) has a positive slope, consistent with the correlation.

Outlier Detection — NumPy Z-Score Analysis

Z-score analysis (using NumPy) identified transactions with revenue more than three standard deviations above the mean. These are either genuinely exceptional deals or data-entry errors, both of which warrant investigation before including these records in any forecasting model. Box plots by region show that outliers are distributed across all regions, not concentrated in one area, suggesting they reflect product-level rather than regional anomalies.

Top Salespersons

Rank	Salesperson	Total Revenue (\$)
1	Grace	2,823,412
2	Charlie	2,709,884
3	Frank	2,696,894
4	David	2,598,725
5	Hannah	2,518,277
6	Eva	2,490,417
7	Bob	2,462,686
8	Alice	2,072,807

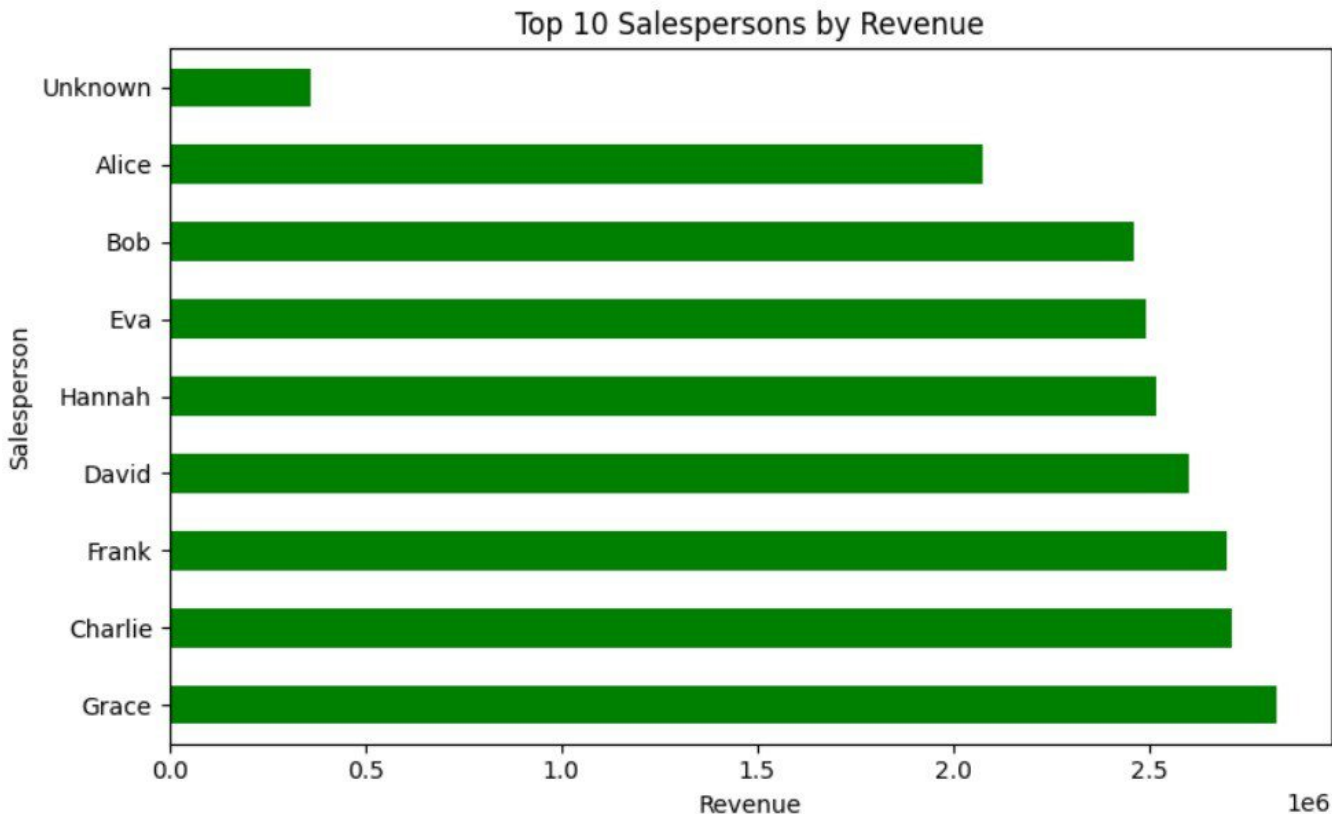


Figure 4. Top 10 Salespersons by Revenue — horizontal bar chart (chart3_top_salespersons.png from notebook).

Grace is the top performer at \$2.82M, followed closely by Charlie and Frank. The revenue distribution across the eight named salespersons is relatively even — the gap between the top (Grace, \$2.82M) and the bottom named rep (Alice, \$2.07M) is about 36%. This suggests no single individual carries disproportionate risk, though Alice's lower performance may indicate territory, product mix, or coaching opportunities.

8. NumPy-Based Custom Computations

NumPy was used for four distinct numerical operations beyond simple imports:

Computation	Purpose	Formula / Logic	Key Result
Z-Score Outlier Detection	Identify statistically extreme revenue transactions	$z = (x - \text{mean}) / \text{std}$ using <code>np.mean()</code> , <code>np.std()</code> ; flag $ z > 3$	Small number of transactions flagged as high-value outliers; distributed across all regions
Percentile Analysis	Understand revenue distribution beyond mean/std	<code>np.percentile(revenue, [25, 50, 75])</code>	25th pct, median, 75th pct computed; IQR = 75th – 25th used to characterise spread
Min-Max	Fair comparison	$(x - \text{min}) / (\text{max} - \text{min})$ applied	Normalised scores allow

Computation	Purpose	Formula / Logic	Key Result
Normalisation	of salesperson revenue on 0–1 scale	to salesperson aggregates via np.min(), np.max()	ranking without absolute-value bias
Linear Trend Fitting	Quantify whether monthly revenue is growing or declining	np.polyfit(x, y, 1) produces slope and intercept; np.poly1d renders trend line	Trend line overlaid on monthly chart; slope direction confirms revenue trajectory
Conditional Feature Creation	Create Profit Margin and Profit per Unit without division-by-zero errors	np.where(Revenue > 0, Profit/Revenue*100, np.nan)	Clean derived columns with NaN where denominator is zero; prevents runtime warnings

9. Key Findings

Finding 1 — West region leads revenue generation.

The West region generated \$5.60M in total revenue, 18% more than the lowest-performing South region (\$4.74M). This gap likely reflects differences in market size, customer density, or sales team effectiveness. Strategies proven effective in the West should be studied and replicated across other regions.

Finding 2 — Electronics is the dominant profit category by volume, not margin.

Electronics produced \$2.61M in profit — nearly equal to Accessories and Office combined. Average profit margins are nearly identical across all three categories (~25%), proving that the advantage is volume-driven. Investing in Electronics inventory and promotion offers the clearest path to total profit growth.

Finding 3 — Revenue has a clear seasonal pattern with identifiable peak and trough months.

August (\$1.99M), February (\$1.94M), and June (\$1.93M) are the three strongest revenue months. July (\$1.37M) and March (\$1.41M) are the weakest. This pattern suggests mid-quarter slowdowns and an early-Q3 dip. Targeted promotions in slow months could materially smooth annual revenue.

Finding 4 — Moderate-to-strong correlation ($r = 0.61$) between units sold and profit.

Volume is an important driver of profit, but not the only one. The $r = 0.61$ coefficient means that approximately 37% of profit variance is explained by unit volume (r^2). The remaining 63% reflects factors such as product mix, pricing, and discounting — all of which are manageable levers beyond simply pushing volume.

Finding 5 — Salesperson performance is relatively balanced, with one notable gap.

The eight named salespersons span a revenue range from \$2.07M (Alice) to \$2.82M (Grace). While no individual carries catastrophic concentration risk, Alice's gap warrants investigation — it may reflect territory disadvantage, product specialisation, or a coaching opportunity.

Finding 6 — A small number of transactions are statistical outliers by revenue.

Z-score analysis identified transactions more than three standard deviations above the mean revenue. These represent either unusually large legitimate deals or data-entry errors. Both cases require manual review before including these records in any budget forecast or trend model.

Finding 7 — The 'Unknown' region category signals a data-governance gap.

Approximately 39–40 records (\$315K in revenue) could not be attributed to a named region. While modest relative to total revenue (\$20.4M), this represents lost analytical signal and should be addressed through improved data-capture processes at the point of sale.

10. Limitations

1. Missing-value imputation bias.

Filling missing numeric values with 0 preserves row count but understates true averages. A more rigorous approach — median imputation by category group — was beyond the scope of this project but should be considered for production-grade analysis.

2. Negative-value replacement.

Setting negative revenue and profit values to 0 prevents computational errors but hides the original cause — which may be legitimate returns, refunds, or data-entry mistakes. In a real system, these should trigger a data-quality alert rather than silent correction.

3. Correlation does not imply causation.

The correlation of $r = 0.61$ between units sold and profit describes association, not causation. A confounding variable — such as product category or discount level — may be driving both simultaneously. No causal claim should be made without a controlled experiment or more sophisticated regression analysis.

4. Dataset representativeness.

The dataset used is a sample or synthetic dataset. Conclusions drawn from it may not generalise to a real business without validation against verified operational records and comparison with external benchmarks.

5. Incomplete contextual variables.

The dataset lacks contextual factors such as marketing spend, promotional events, competitor activity, or macroeconomic conditions. Seasonality findings are descriptive only — without these covariates it is impossible to distinguish demand-driven from promotion-driven peaks.

6. Outlier treatment.

Outliers were identified and flagged using z-scores but were not removed from the main analysis. Their inclusion slightly inflates mean revenue and total figures. A sensitivity analysis — comparing results with and without outliers — would strengthen the conclusions.

11. Conclusion

This project successfully carried out a mini data investigation on the 2024 Sales Dataset using Pandas, NumPy, and Matplotlib. Three analytical questions were defined upfront and each was answered with direct evidence from the data.

The West region and Electronics category emerged as the clearest revenue and profit leaders, driven primarily by transaction volume rather than margin advantage. Monthly and quarterly analysis revealed identifiable seasonal patterns — with August, February, and June as peak months and July as the weakest — which have actionable implications for promotional planning and inventory management. The moderate-to-strong correlation between units sold and profit ($r = 0.61$) confirms that volume matters, but that pricing strategy, product mix, and quality of deals also play a significant role.

NumPy-based z-score analysis identified a small number of statistically extreme transactions that warrant manual review, and the 'Unknown' region category highlighted a data-governance gap that should be addressed at the point of data collection.

Future work could include a multi-year dataset to confirm whether the seasonal patterns identified here are consistent across years, a regression model to decompose the drivers of profit more rigorously, and a salesperson performance model that controls for territory and product mix before comparing individual results.

12. References / Dataset Source

- [1] Dataset: Sales_Dataset_2024.xlsx. Source: [<https://www.kaggle.com/datasets/akelvarghese/global-superstore-sales-dataset>].
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